

Skills Summary

Putting Small Shapes Together to Make Bigger Shapes

Use this reference while completing your worksheet or when studying.

Skill 1 — Counting the pieces that cover a shape

Rule: when the pieces are laid in **equal groups** (rows, or one small shape repeated inside a bigger one), do not count one piece at a time.

Count what is in one group, then count the groups, and skip-count. If a build has layers (triangles inside rhombuses inside a hexagon), start with the smallest piece and work up one layer at a time.

Example: A rectangle is covered with square tiles: 3 rows, and each row holds 4 tiles. How many tiles?

Step 1. Every row holds the same number of tiles, so these are equal groups — count one row, then count the rows.

Step 2. One row is 4 tiles and there are 3 rows: $4 + 4 + 4 = 12$, or $3 \times 4 = 12$. $\Rightarrow$ **12** tiles

Try it: A rectangle is covered with square tiles: 2 rows, and each row holds 5 tiles. How many tiles?

check: 10

Skill 2 — Naming the shape the pieces make

Rule: pieces must touch along **whole sides**, with no gaps and nothing on top of anything.

To name the new shape, **trace the outline** and ignore the lines inside the build. Count the straight sides and the corners of that outline: three straight sides names a triangle; four straight sides with square corners names a rectangle.

Example: Two identical right triangles are joined so that their longest sides touch exactly. Name the new shape.

Step 1. The pieces touch along a whole side, so the line between them is inside the build — trace only the outline.

Step 2. The outline has four straight sides and four square corners, and one pair of sides is longer than the other pair. $\Rightarrow$ rectangle

Try it: Two identical red trapezoids are joined so that their longest sides touch exactly. Trace the outline and name the new shape.

check: hexagon

Skill 3 — Comparing two ways to cover a shape

Rule: when *the same* big shape is covered two different ways, compare the two **counts of pieces** — never how big one piece looks.

Write the two counts and put $>$ or $<$ between them. The bigger count is the covering that used more pieces. **Smaller pieces always means more of them.**

Example: One rectangle is covered by 4 square tiles. The same rectangle is covered by 8 triangle tiles. Which covering uses more pieces?

Step 1. Both coverings fill the same rectangle, so the only thing to compare is how many pieces each one took.

Step 2. Four pieces against eight pieces, and 4 is less than 8. $\Rightarrow 4 < 8$, so the triangle covering uses more pieces

Try it: One hexagon is covered by 2 red trapezoids. The same hexagon is covered by 6 green triangles. Which covering uses more pieces?

check: $2 < 6$

(!) Watch out: “Bigger pieces” does not mean “more pieces”. The bigger the piece, the *fewer* of them it takes to cover the same shape. Also check that the pieces really touch along whole sides — a build with a gap in it has not covered anything.